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- **Problem Statement: IOT-BASED ELECTRICAL MUSCLE STIMULATION AND REAL-TIME MONITORING SYSTEM FOR PARALYSIS REHABILITATION**
- **Theme: Health & Wellbeing**
- **Team Name (Registered on portal): ELECTROCARE**
- **Team ID(Registered on portal): HSC|TN|01080**
- **State/UT: Tamilnadu**

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Problem – Proposed Solution

PROBLEM

PROBLEM

Due to decreased muscle activation, paralyzed people may have muscle atrophy and weakness.

PROBLEM

Conventional rehabilitation methods might not provide remote access or constant monitoring.

PROBLEM

Manual observation makes it difficult to monitor patients regularly

UNIQUENESS

Real-Time Monitoring

Tracks muscle activity, temperature, heart rate and SpO_2.

Remote Patient Monitoring

Enables remote access to patient monitoring data.

Smart Safety Control

Combine abnormal-reading alerts with programmed temperature control.

OUR IDEA

INNOVATION

- The ESP32 monitors vital signs and muscular activity by combining EMG, DS18B20, and MAX30100 sensors.
- Sensor readings are communicated over an IoT platform and seen locally

SUSTAINABILITY

- Caregivers can view patient readings from a distance with the aid of remote access.
- Automatic alerts help identify abnormal sensor measurements..

SAFETY AND SCALABILITY

- LED and buzzer provide visual and audio warnings.
- A relay-controlled Peltier module supports localised temperature adjustment

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Impact Potential

Hack for Social Cause

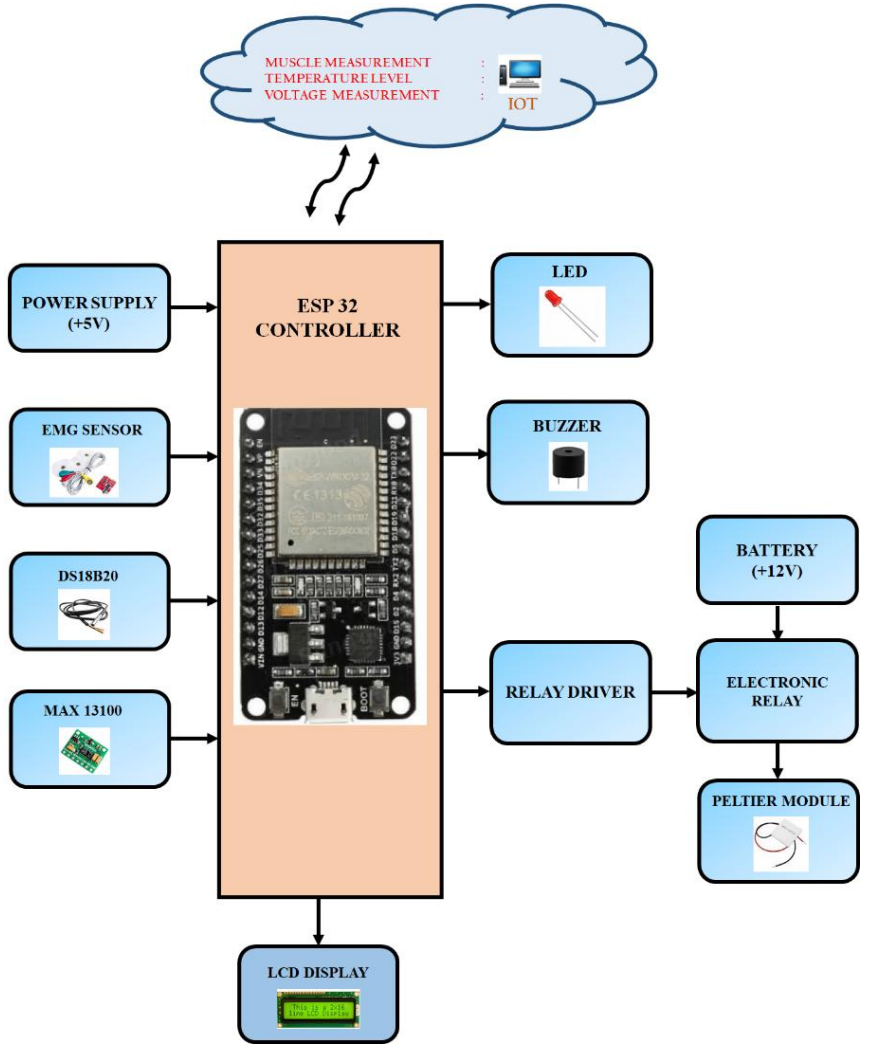
Hardware: ESP32, EMG Sensor, DS18B20 Temperature Sensor, MAX30100 Sensor, LCD Display, LED, Buzzer, Relay Driver, Electronic Relay and Peltier Module.

Programming Language: Embedded C/C++ using Arduino IDE.

IoT Platform: Blynk or ThingSpeak for remote patient monitoring.

Communication: Wi-Fi for wireless data transmission.

Software Tools: Arduino IDE for programming and circuit simulation tools for prototype testing.



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Feasibility and Risk Analysis

Hack for Social Cause

ANALYSIS

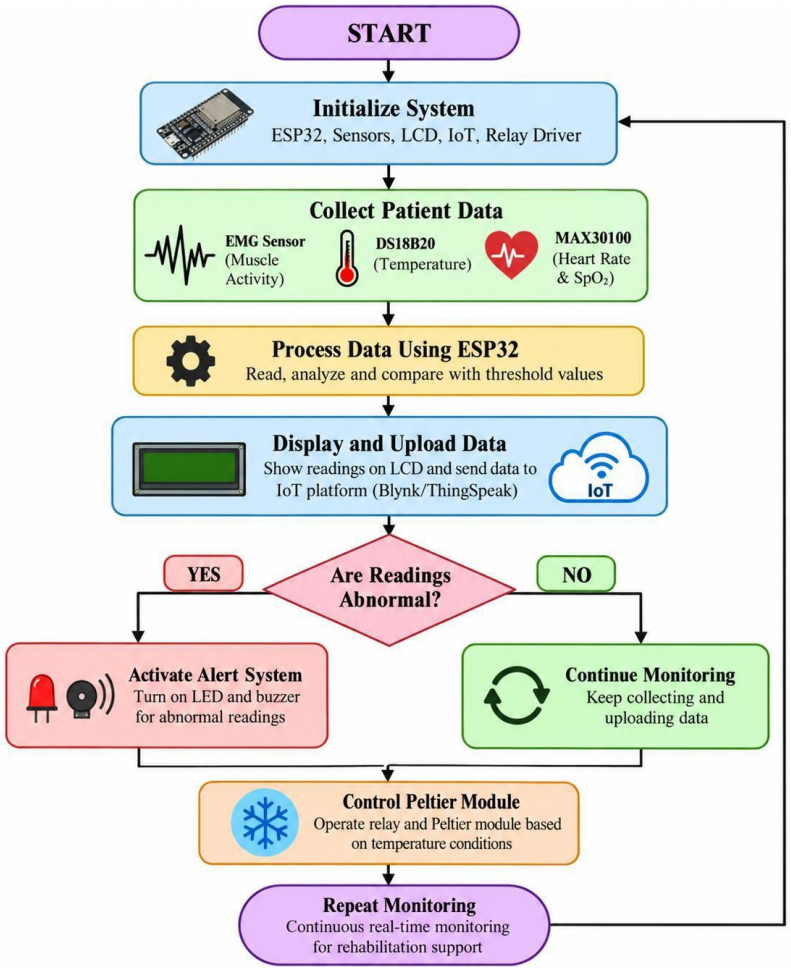
- Using sensors attached to the ESP32, the system tracks blood oxygen levels, heart rate, body temperature, and muscle activity.
- The LCD shows readings, the alarm system alerts caregivers to abnormal values, and IoT connectivity allows for remote patient monitoring.

CHALLENGES

- The accuracy of monitoring may be impacted by sensor values that change as a result of movement, electrical noise, or incorrect connections.
- While improper temperature settings can compromise the Peltier module's safe operation, Wi-Fi outages may cause remote data updates to be delayed

STRATEGIES

- Proper sensor connections, calibration where applicable and individual component testing help improve measurement reliability and system performance.
- Reliable IoT configuration, predefined alert thresholds and temperature safety limits help ensure effective monitoring and controlled operation.



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Team

Team Composition

1. Team Members

- Sandhiya D
- Sharmila V
- Gopika S

2. Roles

- **Sandhiya D:** Hardware components, circuit design, and sensor integration. **Sharmila V:** Project coordination, research, and presentation..
- **Gopika S:** IoT monitoring, documentation, and testing.

3. Challenges

- Understanding EMS technology and its applications.
- Integrating sensors with the ESP32 microcontroller.
- Developing a reliable IoT-based patient monitoring system.
- Ensuring safety and reliable data monitoring.

4. Our Journey

- Identified the problem of muscle atrophy in paralyzed patients.
- Researched existing rehabilitation and monitoring systems.
- Designed an IoT-based EMS and patient monitoring prototype.
- Worked together to develop and improve our project.

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References

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3. R. Merletti, L. R. Lo Conte, and C. Orizio, “Indices of Muscle Fatigue,” Journal of Electromyography and Kinesiology, vol. 1, pp. 20–23, 1991.

4. M. L. Rahman, M. J. Alam, N. Rashid, L. H. Thithy, A. U. Ahmed, and M. T. Arafat, “IoT Based Cost Efficient Muscular Stimulator for Biomedical Application,” in Proceedings of the 2020 IEEE Region 10 Symposium (TENSYMP), Dhaka, Bangladesh, June 2020, pp. 634–637.

5. Y. J. Fan, Y. H. Yin, L. D. Xu, Y. Zeng, and F. Wu, “IoT-Based Smart Rehabilitation System,” IEEE Transactions on Industrial Informatics, vol. 10, no. 2, pp. 1101–1102, 2014.